

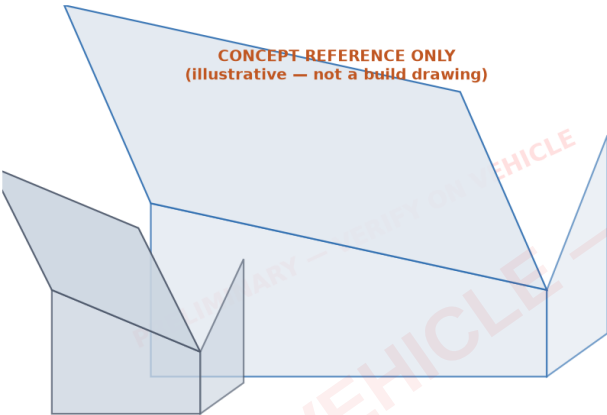
TAMARAW CAMPER

Phase-1 MVP — Dwelling Shell Engineering Specification

PH Toyota Tamaraw LWB Dropside → Weather-tight Dwelling Shell
NO interior fit-out in Phase-1 scope

PRELIMINARY — VERIFY ON VEHICLE — NOT LTO-CERTIFIED

FIG G — CONCEPT MASSING (generated reference)



PH Toyota Tamaraw LWB dropside → dwelling shell Phase-1 MVP

Figure G — Concept massing (illustrative reference only)

Document	Tamaraw_Camper_Spec_v1
Revision	v1.0 — 2026-09-23 (Asia/Shanghai)
Vehicle base	Toyota Tamaraw LWB Dropside (public press dims; verify on VIN)
Scope	Shell structure, envelope, underbody battery cradle, utility apertures
Out of scope	Cabinets, beds, plumbing fixtures, kitchen, finishes, LTO homologation
Materials policy	Mild steel tube / GI sheet / mineral wool / foil — NO plastics/resins/composites
Status	Engineering draft for fabricator discussion — not a certified build package

Five key numbers: Bed 2647×1711 mm · WB 3085 mm · Shell max W 2000 mm · Clear H 1900–2000 / Overall H ≤2500 mm · Battery ~43–45 kWh @48V (7-day example)

1. Document Control & Contents

This specification defines the Phase-1 MVP **dwelling shell** for a Philippine-market Toyota Tamaraw long-wheelbase (LWB) dropside. Phase-1 delivers a weather-tight, structurally tied volume with defined apertures and an underbody battery envelope. Interior habitable fit-out is explicitly excluded.

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1.1 Revision history

Rev	Date	Notes
v1.0	2026-09-23	Initial Phase-1 MVP shell spec; public-press vehicle dims; transparent battery math

1.2 Distribution & liability

Issued for design review and fabricator quotation only. Dimensions from public press must be verified on the actual VIN/vehicle before cutting steel. This document is **not** an LTO (Land Transportation Office) certification package, type-approval submission, or structural PE-stamped drawing set. All structural, electrical, and fire-safety decisions remain the responsibility of the builder and licensed professionals.

2. Scope, Exclusions & Assumptions

2.1 In scope (Phase-1 MVP shell)

- Mild-steel tube primary frame sized to LWB bed footprint (~2647 × 1711 mm clear bed).
- GI sheet skin, mineral-wool cavity fill, foil vapor/radiation barrier; exterior heat-reflective paint (local equivalents).
- Weather-tight shell with defined entry, rear doors, and two utility compartment **openings** (tool + laundry) — apertures and hinged cover panels only.
- Chassis tie-in strategy preferring clamps/cradles/crossmember interfaces — **no undirected frame-rail drilling**.
- Underbody battery enclosure envelope with vent path, fire separation, and mechanical isolation from exhaust/fuel.
- Drawing set: cover, elevations, plan, wall section, underbody, expansion envelope.

2.2 Explicitly out of scope

- Interior fit-out: beds, galley, wet bath, cabinets, furniture, flooring finishes.
- Plumbing fixtures, tanks (beyond reserved interface notes), HVAC equipment selection.
- Chassis extension / stretcher; SWB variant; 4×4 conversions.
- Plastics, FRP, resins, sandwich composites, XPS/EPS foam boards.
- LTO registration change, weight certification, emissions, or roadworthiness homologation.
- Solar array structural design beyond roof load allowance note; inverter/charger BOM finalization.

2.3 Assumptions

- Host vehicle is stock LWB dropside; payload ~1 t order-of-magnitude (press: ~1025–1125 kg depending on variant) — confirm GVWR vs. shell + battery + occupants.
- No chassis length extension preferred; shell length ≤ bed length.
- Phase-1 max exterior shell width 2000 mm; OEM vehicle width ~1800 mm.
- Interior clear height target 1900–2000 mm; overall vehicle+shell height target ≤2500 mm.
- Electrical example is 48 V LiFePO4 for autonomy sizing transparency only — not a purchase order.

3. Vehicle Baseline (Public Press — Verify on VIN)

Sources: Toyota Motor Philippines product pages and Philippine automotive press (CarGuide.PH, Top Gear PH, dealer listings, Zigwheels). Values vary slightly by trim; **measure the actual unit before fabrication**.

Parameter	Public-press value	Spec use / note
Configuration	LWB Dropside	Phase-1 host
Wheelbase (WB)	3085 mm	Primary longitudinal datum
Overall L × W × H	~5300 × 1800 × 1800 mm	OEM cab+dropside; height rises with shell
Bed length × width	2647 × 1711 mm	Shell footprint baseline
Payload (order of mag.)	~1 t (≈1025–1125 kg by trim)	Budget shell+battery+crew carefully
Turning radius (LWB)	~5.4 m	Maneuverability note only
Tread F/R	1510 / 1510 mm	Underbody clearance planning

3.1 Verification gates (mandatory before cut)

- Record VIN, trim, GVWR, curb weight from door sticker / OR/CR.
- Measure bed inner L×W×side-panel height; note stake-pocket / lashing-hook positions.
- Map frame-rail outer faces, crossmember locations, exhaust routing, fuel tank, spare.
- Confirm rear overhang and departure angle with proposed battery cradle depth.
- Weigh proposed BOM (steel + wool + GI + battery) against remaining payload.

Until verification is complete, all dimensions in this document remain PRELIMINARY.

4. Shell Envelope & Dimensional Constraints

Constraint	Phase-1 target	Rationale
Shell exterior width (max)	2000 mm	Controlled over-OEM width; Phase-1 hard cap
Shell length	≤ 2647 mm (bed length)	No chassis extension preferred
Interior clear height	1900–2000 mm	Standing room inside shell volume
Overall height (ground → roof)	≤ 2500 mm	Bridge/parking / PH road practicality
Wall build-up (typ.)	~50–80 mm	GI + 50 mm wool + tube + liner
Floor build-up (typ.)	design TBD on bed	Protect OEM bed; drain / seal

FIG F — EXPANSION ENVELOPE | Phase-1 solid; later phases dashed/dotted

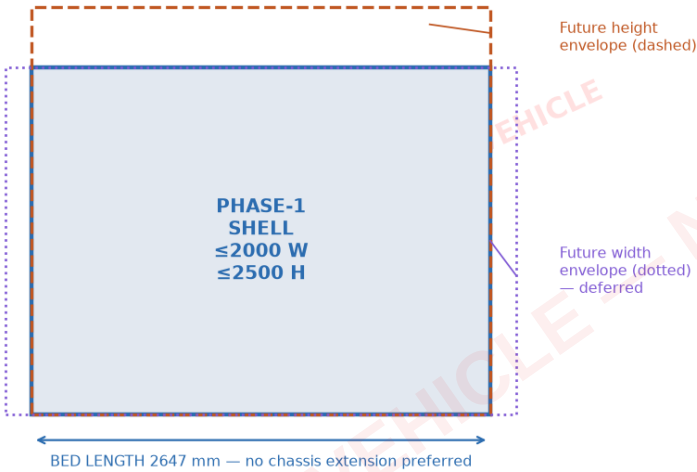


Figure F — Expansion envelope: Phase-1 solid; later width/height dashed/dotted

Width growth beyond 2000 mm and height growth beyond 2500 mm are deferred. Any change requires payload, CG, and regulatory re-check.

5. Elevations, Plan & Wall Section

5.1 Side elevation

FIG A — SIDE ELEVATION (schematic, mm) | Phase-1 dwelling shell on LWB dropside

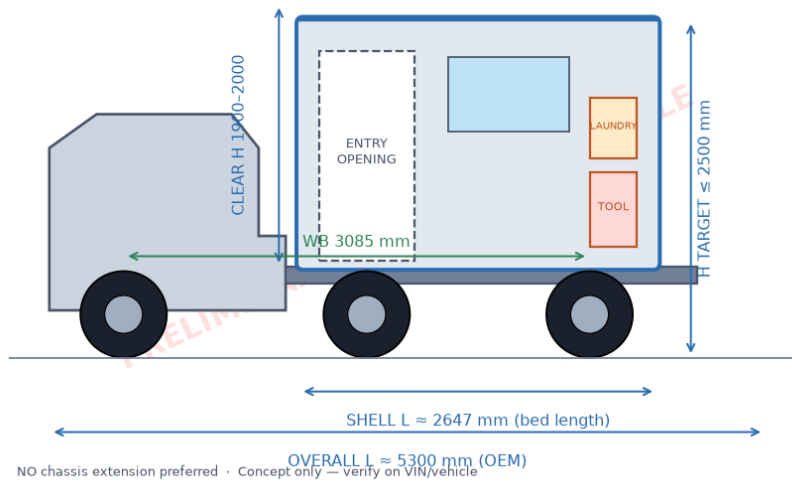


Figure A — Side elevation (schematic). Shell sits on LWB bed; WB 3085 mm shown.

5.2 Rear elevation

FIG B — REAR ELEVATION

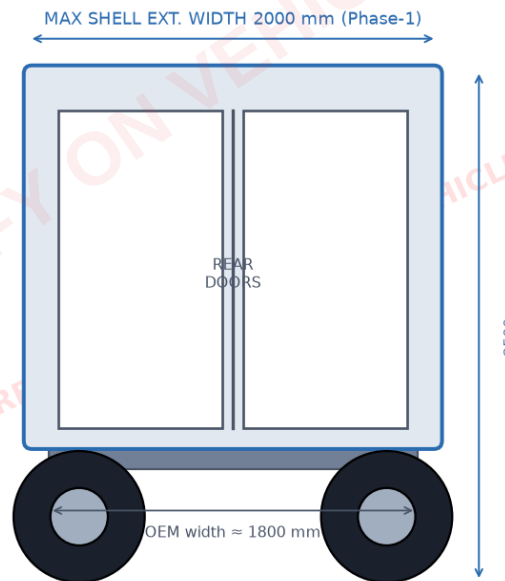


Figure B — Rear elevation. Max shell exterior width 2000 mm; overall H ≤ 2500 mm.

5.3 Plan

FIG C — PLAN (looking down) | Phase-1 shell envelope on LWB bed

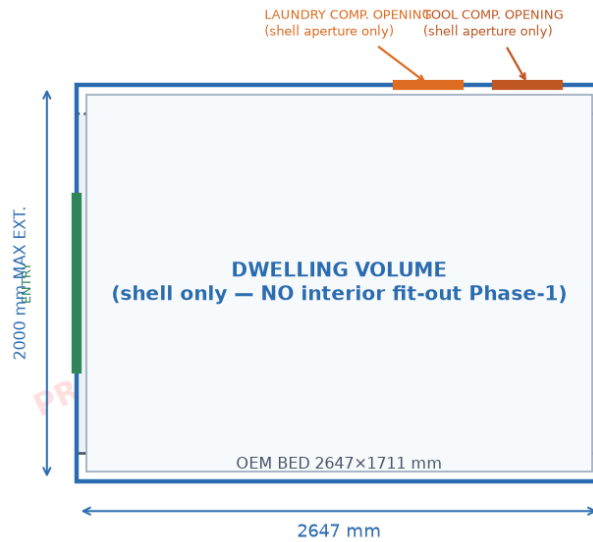


Figure C — Plan. Dwelling volume is shell-only; tool & laundry shown as apertures.

5.4 Typical wall section

FIG D — TYPICAL WALL SECTION (schematic) | exterior → interior

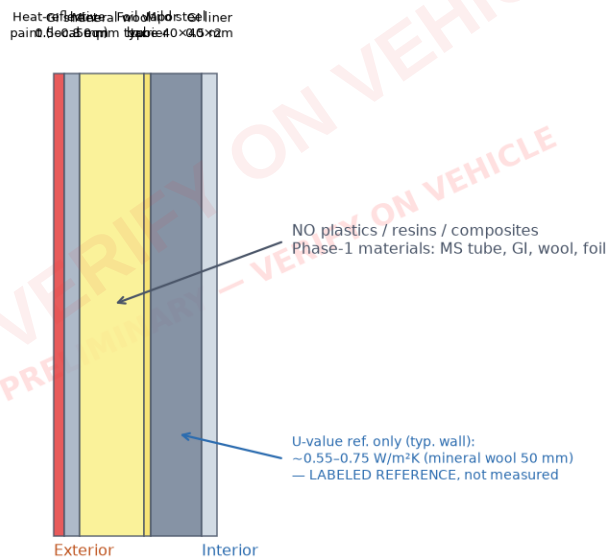


Figure D — Wall section exterior→interior. No plastics/resins/composites.

6. Materials & Thermal References

Phase-1 materials are limited to **mild steel tube, GI sheet, mineral wool, and foil**, plus local heat-reflective exterior paint equivalents. Plastics, resins, and composites are prohibited in the shell build-up.

Item	Spec / size (typ.)	Notes
Primary tube	40×40×2.0 mm MS SHS (alt. 50×50×2)	Main rings / rails; confirm FEA later
Secondary tube	25×25×1.5–2.0 mm MS	Noggins, door bucks, aperture frames
Skin	GI sheet 0.5–0.8 mm	Exterior; seam & seal carefully
Liner	GI sheet 0.5 mm	Interior face of shell (not furniture)
Insulation	Mineral / rock wool 50 mm	Cavity fill; keep dry
Barrier	Aluminum foil / kraft-foil	Vapor / radiant layer
Paint	Local heat-reflective enamel/acrylic equiv.	Exterior solar gain control
Fasteners	M6–M10 Gr8.8 zinc; SS where wet	See §9
Hinges	Heavy butt / strap, steel	Entry, rear, utility doors

6.1 Thermal numbers — labeled references only

The following are **typical literature / catalog order-of-magnitude values** for communication with fabricators. They are **not** project-measured U-values and must not be used for energy-code compliance claims.

Property	Reference value	Label
Mineral wool λ (typ.)	~0.035–0.040 W/m·K	REFERENCE ONLY
50 mm wool R (approx.)	~1.25–1.4 m ² K/W	REFERENCE ONLY
Wall U (schematic build-up)	~0.55–0.75 W/m ² K	REFERENCE ONLY — not tested
GI emissivity (unpainted)	high (metal)	Paint lowers solar absorptance
Heat-reflective paint	reduces roof solar gain (qualitative)	Local product TBD

7. Underbody Battery & 7-Day Autonomy Math

7.1 Transparent autonomy example

Design goal: approximately **7 days** of electrical autonomy at a continuous average load of **~180 W** (lights, fans, devices, controls — Phase-1 shell-level loads; not a full camper fit-out load profile).

- Daily energy: $180\text{ W} \times 24\text{ h} = 4.32\text{ kWh/day}$
- 7-day usable need: $4.32 \times 7 = 30.24\text{ kWh usable}$
- Assume ~70% usable fraction of nameplate (DoD + reserve + inefficiencies, example): $30.24 / 0.70 \approx 43.2\text{ kWh nameplate}$
- Rounding / contingency band: **~43–45 kWh @ 48 V LiFePO4** example chemistry
- Current (order): $45\,000\text{ Wh} / 48\text{ V} \approx 940\text{ Ah}$ pack class (example — vendor modules TBD)

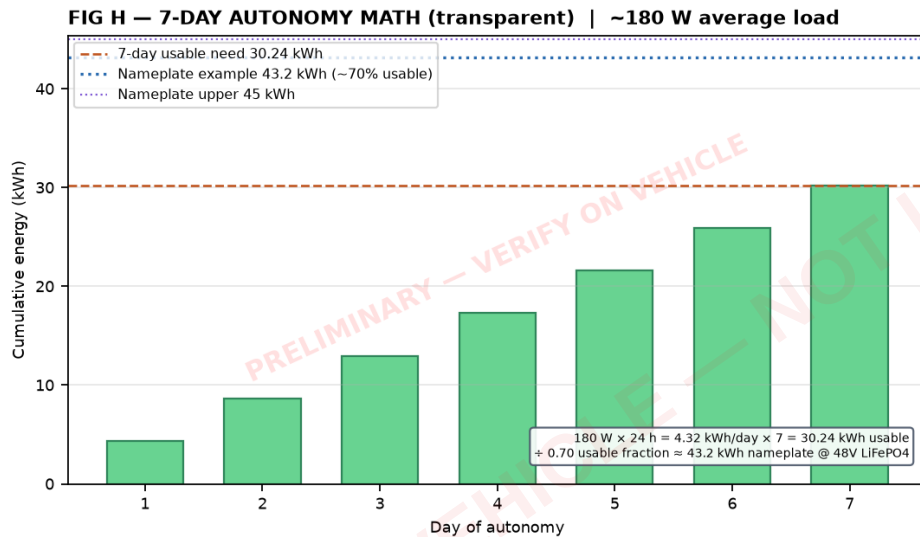
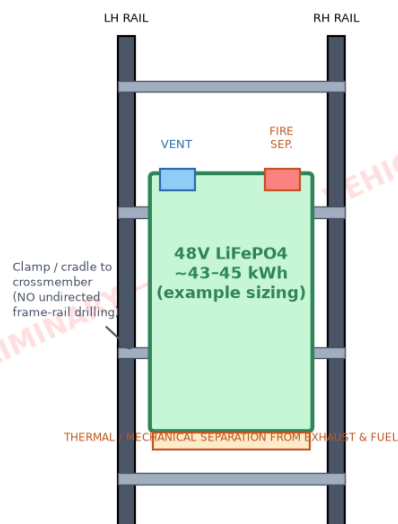


Figure H — Cumulative energy vs. day; usable need vs. nameplate examples.

7.2 Underbody placement — vent / fire / separation

FIG E — UNDERBODY BATTERY ENVELOPE (plan schematic)



Autonomy example: $180\text{ W avg} \times 24\text{ h} \times 7\text{ d} = 30.24\text{ kWh usable} \rightarrow \sim 43\text{--}45\text{ kWh nameplate @ } \sim 70\% \text{ usable depth}$

Figure E — Underbody battery envelope between frame rails (schematic).

7.3 Safety & integration requirements (shell-level)

- **Ventilation:** dedicated vent path from enclosure to outside air; prevent battery bay from being a sealed heat trap.
- **Fire separation:** steel enclosure; keep clearances from exhaust; provide thermal shield where proximity is unavoidable; dry chemical / Class D considerations per pack vendor.
- **Mechanical separation:** isolate from fuel tank, brake lines, and moving driveline; use cradle that does not depend on random rail holes.
- **Ingress:** IP-rated gland for HV/comms cables; drip loops; no water ponding on cover.
- **Service:** pack removable with vehicle on stands; disconnect sequence placarded.
- **Mass:** LiFePO4 ~43–45 kWh may approach ~300–400+ kg depending on modules/enclosure — confirm against payload and rear axle rating.

Electrical design beyond envelope (BMS, fusing, inverter, charger, solar) is Phase-1 **interface-reserved** only and requires a separate electrical package.

8. Utility Openings (Tool + Laundry) — Shell Apertures Only

Phase-1 provides **openings and hinged cover panels** only. Interior cabinets, baskets, plumbing, and appliances are out of scope.

Opening	Nominal clear (mm)	Location (intent)	Hardware
Tool compartment	350 W × 550 H	Starboard aft lower	Strap hinge + padlock hasp
Laundry compartment	350 W × 450 H	Starboard aft mid	Strap hinge + compression latch
Entry door	700 W × 1550 H (min)	Port / forward shell	Butt hinges + seal
Rear barn doors	pair ~800 W each × 1550 H	Aft wall	Butt/strap + flush bolts

- Frame each aperture with secondary tube bucks continuous to primary rings.
- Provide drip edge / weather seal; drain away from battery and chassis electrics.
- Do not cut primary longitudinal rails for utility doors — penetrate skin/secondary only with designed bucks.
- Label interiors of covers: TOOL / LAUNDRY — SHELL OPENING ONLY (PHASE-1).

9. Fabrication: Cut Lists, Welds, Fasteners, Chassis Tie-ins

9.1 Indicative cut list (adjust after VIN survey)

Mark	Member	Size	Qty (indic.)	Length ea. (mm)
PR-1	Primary ring upright	40×40×2 SHS	8	~1950–2050
PR-2	Primary ring top/bottom	40×40×2 SHS	8	~1900–2000 (W)
PL-1	Longitudinal top rail	40×40×2 SHS	2	≤2647
PL-2	Longitudinal bottom rail	40×40×2 SHS	2	≤2647
XG-1	Roof/floor cross	40×40×2 / 25×25	10–14	~1900–2000
DR-1	Door/aperture bucks	25×25×2	lot	per opening
SK-1	Exterior GI panels	0.5–0.8 mm	lot	nest from sheets
LN-1	Interior GI liner	0.5 mm	lot	nest from sheets
INS-1	Mineral wool batts	50 mm	lot	wall+roof area
FL-1	Foil barrier	roll	1–2	wall+roof area

Cut list is indicative for quotation. Final nesting after field measure. Mass estimate must be updated before ordering the battery pack.

9.2 Welds

- Process: SMAW / GMAW (MIG) on clean MS; typical fillet 3–4 mm on 2 mm tube (verify WPS).
- Full-penetration or double-sided fillets at primary ring corners; stitch welds acceptable on non-critical skin lands with anti-corrosion seal.
- Dress sharp edges; zinc-rich primer on all heat-affected zones after weld clean-up.
- Do not weld to OEM frame rails. Shell welds are on the shell assembly only.
- NDT: visual 100%; dye-penetrant on critical corner samples as fabricator QA.

9.3 Fasteners & hinges

- Structural bolted joints: M8–M10 Grade 8.8, Nyloc or prevailing-torque nuts, washers both sides.
- Skin: self-drilling screws into tube lands or rivets with sealant; spacing ~150–200 mm at edges.
- Hinges: steel butt or strap, min. 3 per tall door; greaseable pins preferred.
- Latches: compression for utility; slam + deadbolt capability for entry; padlock hasps where noted.
- Dissimilar metal: isolate SS fasteners from GI with washers / compound where practical.

9.4 Chassis tie-ins — no undirected frame-rail drilling

Attachment philosophy: transfer shell loads into the OEM bed structure and/or crossmembers via **designed clamps, saddles, and through-bolted bed interfaces**. Random or exploratory drilling of chassis frame rails is prohibited.

Interface	Method	Forbidden
Shell → bed floor	Through-bolt at reinforced lands / existing stake or bed structure	Undersized self-tappers into thin bed only
Shell → side panels	Remove or notch dropside as designed; weld shell to its own sill	Welding directly to unknown OEM high-strength zones without survey
Battery cradle → chassis	Clamp/saddle to crossmembers; optional OEM hole reuse if documented	Undirected drilling of longitudinal frame rails
Electrical bond	Dedicated ground strap to OEM ground points	Reliance on painted hinge only

- If a new hole in a rail is ever proposed, it requires written structural rationale, location drawing, corrosion protection, and owner approval — treat as a controlled deviation, not a field habit.
- Preserve OEM brake/fuel/EVAP clearances; mock-up cradle in cardboard/ply before steel.
- After install: check torque, re-torque after 100 km, inspect weld and clamp points each service.

10. Expansion Envelope & Phase-2 Hooks

Phase-1 freezes width ≤ 2000 mm and overall height ≤ 2500 mm. The expansion figure (§4 / Fig F) shows deferred envelopes only.

- Roof: leave flat GI lands and primary tube capable of later solar rack (mass TBD) — no array in Phase-1 BOM.
- Interior: do not pre-install furniture hardpoints that lock layout; optional blank tube noggins only.
- Electrical: reserve conduit path from underbody pack to a future distribution panel location.
- Water: optional capped wall penetration bosses (steel) for later plumbing — capped and sealed in Phase-1.
- Width/height increases require new payload/CG analysis and are out of Phase-1 change control without revision.

10.1 Mass & CG caution

Shell steel + wool + GI may be on the order of a few hundred kilograms depending on member sizes; battery example may add several hundred more. Combined with two occupants and cargo, the ~1 t payload band can be consumed quickly. Maintain a running mass budget spreadsheet before procurement.

11. Risks, Verification Checklist & Next Actions

Risk	Mitigation
Press dims ≠ actual bed	Full measure + photo survey before cut list release
Payload overrun (shell+battery)	Mass budget; consider smaller pack or staged battery
Unauthorized rail drilling	Clamp-first policy; deviation log
Water ingress at apertures	Drip edges, seals, hose test
Battery thermal / fire	Steel enclosure, vent, separation, vendor BMS
LTO / registration mismatch	Keep as dropside conversion discussion with counsel; this PDF ≠ certification
Corrosion (PH climate)	Primer, seam sealer, drain holes, inspect annually

11.1 Pre-fabrication checklist

- VIN / GVWR / curb recorded
- Bed L×W×H measured; stake/lashing map
- Exhaust, tank, spare, crossmember photos
- Mass budget v1 signed off
- Cut list revised to field dims
- Clamp mock-up for battery cradle
- Hose-test plan written
- Owner acknowledges NOT LTO-CERTIFIED status

11.2 Immediate next actions

- Schedule vehicle survey day; freeze dimensions.
- Select tube supplier and GI gauge; request mill certs.
- Shortlist 48 V LiFePO4 module vendors for ~43–45 kWh class; get mass & outline drawings.
- Engage fabricator for quotation against this v1; iterate cut list.
- Open Phase-1.1 electrical interface note (separate doc).

Appendix A — Concept References & Drawing Index

Concept imagery in this package is **illustrative only** and must not be scaled for fabrication. Engineering control is via the dimensioned schematic figures and tables herein, subject to VIN verification.

FIG G — CONCEPT MASSING (generated reference)

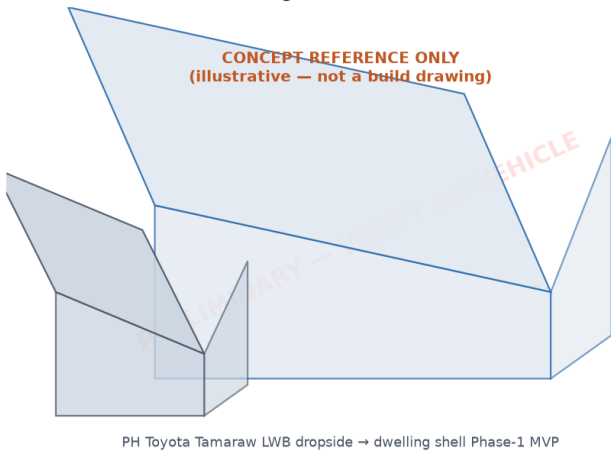


Figure G — Concept massing reference (generated).

Fig	Title	File
A	Side elevation	figures/01_side_elevation.png
B	Rear elevation	figures/02_rear_elevation.png
C	Plan	figures/03_plan.png
D	Wall section	figures/04_wall_section.png
E	Underbody battery	figures/05_underbody.png
F	Expansion envelope	figures/06_expansion.png
G	Concept massing	figures/07_concept.png
H	Autonomy math chart	figures/08_autonomy.png

A.1 Five key numbers (summary)

#	Key number	Value
1	LWB bed (L × W)	2647 × 1711 mm
2	Wheelbase	3085 mm
3	Max shell exterior width (Phase-1)	2000 mm
4	Interior clear H / overall H target	1900–2000 mm / ≤2500 mm
5	Battery example (7-day @ ~180 W avg)	~43–45 kWh @ 48 V LiFePO4

END OF SPECIFICATION — Tamaraw_Camper_Spec_v1
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